

AP CALCULUS AB – UNIT 2

2.5 Product and Quotient Rules



Name: _____ Class: _____ Date: _____

Lesson Overview and Learning Objectives

- Differentiate products of differentiable functions.
- Use values and derivative values from tables or graphs in the Product Rule.
- Choose between expanding first and applying the Product Rule.
- Differentiate quotients of differentiable functions.
- Apply the Quotient Rule to symbolic, tabular, and contextual representations.
- Respect denominator restrictions and simplify strategically.

Keys: Essential Knowledge and Vocabulary

Product Rule: Product Rule

$$(fg)' = f'g + fg'$$

Product Rule: At a point

$$(fg)'(a) = f'(a)g(a) + f(a)g'(a).$$

Product Rule: Three factors

$$(fgh)' = f'gh + fg'h + fgh'.$$

Product Rule: Common error

In general, $(fg)' \neq f'g'$.

Quotient Rule: Quotient Rule

$$\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}, \text{ where } g \neq 0.$$

Quotient Rule: Memory structure

Low times derivative of high minus high times derivative of low, over low squared.

Quotient Rule: At a point

All four values $f(a)$, $f'(a)$, $g(a)$, $g'(a)$ are needed unless a term vanishes.

Quotient Rule: Domain

The original denominator and the squared denominator in the derivative cannot be zero.

Explanation, Strategy, and Common Errors

- **Product Rule.** Name the factors before differentiating. Keep one factor unchanged while differentiating the other, then reverse roles and add.
- **Product Rule.** If both factors are simple polynomials, expansion can be faster and provides a useful check. If factors have meaningful structure or include transcendental functions, the Product Rule is usually cleaner.
- **Product Rule.** Table problems require four values at the target: f , f' , g , and g' .
- **Quotient Rule.** Write the denominator squared first, then construct the ordered numerator. The minus sign is a frequent source of errors.
- **Quotient Rule.** Before using the rule, check whether algebraic simplification turns the quotient into a sum of powers. Simplifying can be faster, but domain restrictions from canceled factors must still be remembered.
- **Quotient Rule.** A horizontal tangent occurs where the derivative numerator is zero while the original

denominator is nonzero.

Solved Example: Product Rule - Symbolic product

Differentiate $f(x) = x^3 \sin x$.

Answer and Reasoning

$$f'(x) = 3x^2 \sin x + x^3 \cos x.$$

Solved Example: Product Rule - Table values

If $u(2) = 3$, $u'(2) = -1$, $v(2) = 5$, and $v'(2) = 4$, find $(uv)'(2)$.

Answer and Reasoning

$$(-1)(5) + (3)(4) = 7.$$

Solved Example: Product Rule - Expand or use the rule

Differentiate $(x^2 + 1)(x - 4)$.

Answer and Reasoning

$$\text{Product Rule: } 2x(x - 4) + (x^2 + 1) = 3x^2 - 8x + 1.$$

Solved Example: Quotient Rule - Symbolic quotient

Differentiate $f(x) = \frac{x^2+1}{x-3}$.

Answer and Reasoning

$$f'(x) = \frac{2x(x-3)-(x^2+1)}{(x-3)^2} = \frac{x^2-6x-1}{(x-3)^2}.$$

Solved Example: Quotient Rule - Table values

If $u(1) = 4$, $u'(1) = 2$, $v(1) = -1$, and $v'(1) = 3$, find $(u/v)'(1)$.

Answer and Reasoning

$$\frac{2(-1)-4(3)}{(-1)^2} = -14.$$

Solved Example: Quotient Rule - Simplify first

Differentiate $\frac{x^4-2x}{x^2}$ for $x \neq 0$.

Answer and Reasoning

Rewrite as $x^2 - 2x^{-1}$, so the derivative is $2x + 2x^{-2}$.

Shared Representation / Data

Product Rule

x	-1	0	1	2
$f(x)$	3	-2	4	5
$f'(x)$	-1	3	2	-4
$g(x)$	2	5	-1	3
$g'(x)$	4	-2	6	1

Quotient Rule

x	-1	0	1	2
$f(x)$	3	-2	4	5
$f'(x)$	-1	3	2	-4
$g(x)$	2	5	-1	3
$g'(x)$	4	-2	6	1

AP Exam Language

For every derivative in context, include the sign, units, input value, and a complete verbal interpretation. For every existence claim, state the relevant hypothesis or one-sided comparison.

Leveled Practice

Easy Foundations

Build fluency with the definition, notation, and one-step applications.

1. Product Rule: Differentiate x^2e^x .

2. Product Rule: Differentiate $(x^3 + 1) \sin x$.

3. Product Rule: If $f(1) = 2$, $f'(1) = 5$, $g(1) = -3$, $g'(1) = 4$, find $(fg)'(1)$.

4. Product Rule: Differentiate $(2x - 1)(x^2 + 3)$.

5. Product Rule: Explain why $(x^2 \sin x)'$ is not $2x \cos x$.

6. Quotient Rule: Differentiate $\frac{x+1}{x-1}$.

7. Quotient Rule: Differentiate $\frac{\sin x}{x}$.

8. Quotient Rule: If $f(2) = 3$, $f'(2) = 4$, $g(2) = 5$, $g'(2) = 1$, find $(f/g)'(2)$.

9. Quotient Rule: Differentiate $1/x^5$ without the Quotient Rule.

10. Quotient Rule: State the domain restriction for the derivative of $\frac{e^x}{x+2}$.

Medium Applications

Connect representations and apply the lesson procedure in familiar settings.

11. Product Rule: Differentiate $f(x) = x^4 \ln x$ for $x > 0$.

12. Product Rule: Find the tangent slope of $y = (x^2 + 2)e^x$ at $x = 0$.

13. Product Rule: Using the shared table, find $(fg)'(1)$.

14. Product Rule: Differentiate $x(x - 1)(x + 2)$ using either method.

15. Product Rule: Find all x where the derivative of $x \sin x$ equals zero.

16. Quotient Rule: Differentiate $\frac{e^x}{x^2 + 1}$.

17. Quotient Rule: Using the shared table, find $(f/g)'(0)$.

18. Quotient Rule: Find the tangent line to $y = \frac{x}{x+1}$ at $x = 1$.

19. Quotient Rule: Find the horizontal tangents of $f(x) = \frac{x^2+1}{x}$.

20. Quotient Rule: Differentiate $\frac{\ln x}{x}$ for $x > 0$.

Hard Reasoning

Select a method, justify conclusions, and combine several ideas.

21. Product Rule: Find $(f^2)'(a)$ in terms of $f(a)$ and $f'(a)$ without invoking the Chain Rule.

22. Product Rule: If $f(2) = 0$, $f'(2) = 3$, and $g(2) = 5$, find $(fg)'(2)$ without knowing $g'(2)$.

23. Product Rule: Find the tangent line to $y = x^2 \ln x$ at $x = 1$.

24. Product Rule: Differentiate $(x^2 + 1)(x^{-1} - 2)$ and simplify.

25. Quotient Rule: Find a so the tangent to $f(x) = \frac{x+a}{x-1}$ at $x = 0$ has slope 5.

26. Quotient Rule: If $f(a) = 0$ and $g(a) \neq 0$, simplify $(f/g)'(a)$.

27. Quotient Rule: Find the tangent line to $y = \frac{\sin x}{1+\cos x}$ at $x = \pi/2$.

28. Quotient Rule: Find the derivative of $\frac{x^3-1}{x-1}$ for $x \neq 1$ and explain the value at $x = 1$.

Difficult Analysis

Work through less-direct algebra, subtle hypotheses, and multi-step reasoning.

29. Product Rule: Given $F = fg$, $F(a) = 0$, and $F'(a) \neq 0$, what can be concluded about $f(a)$, $g(a)$?

30. Product Rule: Find constants a, b so $f(x) = (ax + b)e^x$ has $f(0) = 2$ and $f'(0) = 5$.

31. Product Rule: If $f(x)g(x) = 1$ for all x and both are differentiable, derive a relation between their derivatives.

32. Quotient Rule: Find all horizontal tangents of $f(x) = \frac{x^2-4x+5}{x-2}$.

33. Quotient Rule: If $q = f/g$ and $q(a) = 2$, $q'(a) = 0$, $g(a) = 3$, $g'(a) = -1$, find $f(a)$ and $f'(a)$.

34. Quotient Rule: Derive the reciprocal rule $(1/g)' = -g'/g^2$ from the Quotient Rule.

Challenge / AP Extension

Synthesize the topic in unfamiliar, proof-oriented, or parameter-based problems.

35. Product Rule: Prove the Product Rule from the limit definition by adding and subtracting $f(x+h)g(x)$.

36. Product Rule: Find the n th pattern suggested by differentiating a product of four differentiable factors.

37. Quotient Rule: Prove the Quotient Rule by writing $f/g = f \cdot (1/g)$ and using the Product Rule plus the reciprocal rule.

38. Quotient Rule: Find a condition for f/g to have a stationary point at a in terms of relative rates, assuming $f(a), g(a) \neq 0$.

AP-Style Multiple-Choice Practice

Part A: No Calculator

Choose the best answer. Justification is not required on the AP multiple-choice section, but show reasoning while practicing.

1. **Product Rule:** $(fg)' =$

- (A) $f'g'$ (B) $f'g + fg'$
(C) $fg' + g$ (D) $f' + g'$

2. **Product Rule:** If $f(0) = 2$, $f'(0) = 3$, $g(0) = 4$, $g'(0) = -1$, then $(fg)'(0) =$

- (A) 5 (B) 10
(C) 12 (D) 14

3. **Product Rule:** The derivative of xe^x is

- (A) e^x (B) xe^x
(C) $(x+1)e^x$ (D) x

4. **Product Rule:** Which expression is the derivative of $(x^2 + 1)\cos x$?

- (A) $2x\cos x - (x^2 + 1)\sin x$ (B) $2x(-\sin x)$
(C) $(x^2 + 1)\sin x$ (D) $2x\cos x + (x^2 + 1)\sin x$

5. **Quotient Rule:** $\left(\frac{f}{g}\right)' =$

- (A) $\frac{f'g - fg'}{g^2}$ (B) $\frac{f'g + fg'}{g^2}$
(C) $\frac{f'}{g'}$ (D) $\frac{g'f - gf'}{f^2}$

6. **Quotient Rule:** The derivative of $\frac{x}{x+1}$ is

- (A) $\frac{1}{x+1}$ (B) $\frac{1}{(x+1)^2}$
(C) $\frac{2x+1}{(x+1)^2}$ (D) 0

7. **Quotient Rule:** If $g(a) = 0$, which statement is correct about $(f/g)'(a)$?

- (A) It is always zero. (B) The quotient function is not defined at a , so the derivative there is not defined.
(C) It equals $f'(a)/g'(a)$. (D) It exists if $f(a) = 0$.

8. **Quotient Rule:** For $h(x) = \frac{\ln x}{x}$, $h'(e)$ is

- (A) 0 (B) $1/e$
(C) $1/e^2$ (D) $-1/e^2$

Part B: Graphing Calculator Permitted

Use a calculator only for numerical evaluation, graphing, or regression as indicated. Preserve mathematical setup.

9. Product Rule: Using the shared table, $(fg)'(2)$ equals

- | | |
|-----------|----------|
| (A) -17 | (B) -7 |
| (C) 17 | (D) 23 |

10. Product Rule: At $x = 1.4$, a calculator gives $f = 2.8$, $f' = -0.6$, $g = 5.1$, $g' = 1.7$. Then $(fg)'$ is closest to

- | | |
|-------------|-------------|
| (A) -7.82 | (B) -3.06 |
| (C) 1.70 | (D) 4.76 |

11. Quotient Rule: Using the shared table, $(f/g)'(1)$ equals

- | | |
|-----------|-----------|
| (A) -26 | (B) -22 |
| (C) 22 | (D) 26 |

12. Quotient Rule: At $x = 2.5$, $f = 6.2$, $f' = 1.8$, $g = 3.1$, $g' = -0.4$. Then $(f/g)'$ is closest to

- | | |
|------------|------------|
| (A) 0.32 | (B) 0.84 |
| (C) 1.16 | (D) 2.00 |

AP-Style Free-Response Practice

Calculator Permitted

Complete the following independent calculator-permitted problems. Show the mathematical setup and justify conclusions when requested.

(1a) Problem 1: Product Rule. Functions f and g are differentiable and satisfy $f(2) = 3$, $f'(2) = -2$, $g(2) = 4$, and $g'(2) = 5$. Let $H(x) = f(x)g(x)$.

Find $H(2)$. [1 points]

(1b) Find $H'(2)$. [2 points]

(1c) Write the tangent-line approximation to H near $x = 2$. [2 points]

(2a) Problem 2: Quotient Rule. Let $H(x) = \frac{x^2+3}{x-1}$.

Find $H'(x)$. [3 points]

(2b) Find all horizontal tangents. [2 points]

(2c) Write the tangent line at $x = 3$. [2 points]

No Calculator

Complete the following independent non-calculator problems. Show the mathematical setup and justify conclusions when requested.

(1a) Problem 1: Product Rule. The shared table gives values of f , g and their derivatives. Define $P(x) = f(x)g(x)$.

Find $P'(-1)$. [2 points]

(1b) Find the equation of the tangent line to P at $x = -1$. [2 points]

(1c) Use the line to estimate $P(-0.96)$. [2 points]

(2a) Problem 2: Quotient Rule. Using the shared table, define $Q(x) = \frac{f(x)}{g(x)}$.

Find $Q(2)$. [1 points]

(2b) Find $Q'(2)$. [2 points]

(2c) Use the tangent line to estimate $Q(2.03)$. [2 points]

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